HDOC Day 33

Aim

To write a menu-driven C program using separate functions and variable-length 2D integer arrays to verify the associative properties of matrix addition and multiplication, displaying both resulting matrices.

Theory

Matrix addition is associative for matrices of equal dimensions: (A + B) + C = A + (B + C). Matrix multiplication is associative for conformable matrices A (m x n), B (n x p), and C (p x q): (A x B) x C = A x (B x C). Both sides have dimensions m x q. The program calculates and compares both sides element by element.

Algorithm

1. Matrix Addition: Read m and n and matrices A, B, C of size m x n. Calculate T1 = A + B, then L = T1 + C. Calculate T2 = B + C, then R = A + T2. Display L and R and compare all corresponding elements.

2. Matrix Multiplication: Read m, n, p, q and matrices A (m x n), B (n x p), C (p x q). Compute T1 = A x B and L = T1 x C. Compute T2 = B x C and R = A x T2. Display L and R and compare their m x q elements.

3. Menu: Repeat displaying choices 1, 2 and 3. Execute the selected operation using its function; terminate on choice 3. Reject invalid choices and non-positive dimensions.

C Program

```
#include <stdio.h>

void read(int r, int c, int a[r][c]) {
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++) scanf("%d", &a[i][j]);
}
void show(int r, int c, int a[r][c]) {
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) printf("%d ", a[i][j]);
        printf("\n");
    }
}
void add(int r, int c, int a[r][c], int b[r][c],
         int out[r][c]) {
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++) out[i][j] = a[i][j] + b[i][j];
}
void multiply(int r, int k, int c, int a[r][k],
              int b[k][c], int out[r][c]) {
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++) {
            out[i][j] = 0;
            for (int t = 0; t < k; t++)
                out[i][j] += a[i][t] * b[t][j];
        }
}
int equal(int r, int c, int a[r][c], int b[r][c]) {
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            if (a[i][j] != b[i][j]) return 0;
    return 1;
}
void addition(void) {
    int m, n;
    printf("Enter rows and columns: "); scanf("%d%d", &m, &n);
    if (m <= 0 || n <= 0) { puts("Invalid dimensions"); return; }
```

```
    int A[m][n], B[m][n], C[m][n];
    int T[m][n], L[m][n], R[m][n];
    puts("Enter matrix A:"); read(m,n,A);
    puts("Enter matrix B:"); read(m,n,B);
    puts("Enter matrix C:"); read(m,n,C);
    add(m,n,A,B,T); add(m,n,T,C,L);
    add(m,n,B,C,T); add(m,n,A,T,R);
    puts("(A+B)+C:"); show(m,n,L);
    puts("A+(B+C):"); show(m,n,R);
    puts(equal(m,n,L,R) ? "Associative property verified"
                        : "Associative property not verified");
}
void multiplication(void) {
    int m,n,p,q;
    printf("Enter m n p q: "); scanf("%d%d%d%d", &m,&n,&p,&q);
    if (m<=0 || n<=0 || p<=0 || q<=0) {
        puts("Invalid dimensions"); return;
    }
    int A[m][n], B[n][p], C[p][q];
    int AB[m][p], BC[n][q], L[m][q], R[m][q];
    puts("Enter matrix A:"); read(m,n,A);
    puts("Enter matrix B:"); read(n,p,B);
    puts("Enter matrix C:"); read(p,q,C);
    multiply(m,n,p,A,B,AB); multiply(m,p,q,AB,C,L);
    multiply(n,p,q,B,C,BC); multiply(m,n,q,A,BC,R);
    puts("(A*B)*C:"); show(m,q,L);
    puts("A*(B*C):"); show(m,q,R);
    puts(equal(m,q,L,R) ? "Associative property verified"
                        : "Associative property not verified");
}
int main(void) {
    int choice;
    while (1) {
        puts("\n1. Associative addition");
        puts("2. Associative multiplication");
        puts("3. Exit");
        printf("Enter choice: ");
        if (scanf("%d", &choice) != 1) return 0;
        switch (choice) {
            case 1: addition(); break;
            case 2: multiplication(); break;
            case 3: return 0;
            default: puts("Invalid choice");
        }
    }
}
```

Sample Output

Operation 1: Associative Addition

```
Enter choice: 1
Enter rows and columns: 2 2
Enter matrix A:
1 2
3 4
Enter matrix B:
5 6
7 8
Enter matrix C:
1 1
1 1
(A+B)+C:
7 9
11 13
A+(B+C):
7 9
11 13
Associative property verified
```

Operation 2: Associative Multiplication

```
Enter choice: 2
Enter m n p q: 2 2 2 2
Enter matrix A:
1 2
3 4
Enter matrix B:
1 0
0 1
Enter matrix C:
2 1
1 2
(A*B)*C:
4 5
10 11
A*(B*C):
4 5
10 11
Associative property verified
Enter choice: 3
```

Result

The menu-driven C program was written to verify the associative properties of matrix addition and multiplication using separate functions and variable-length 2D arrays. Both sides of each equation produced identical matrices for the sample inputs; hence both properties were verified.